

MCCONNELL AFB, KS, USA

WMO: 724505

Lat: 37.617N

Lon: 97.267W

Elev: 418

StdP: 96.40

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03923

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-10.7	-19.2	0.7	-10.3	-16.9	0.9	-8.1	13.5	2.7	12.2	4.1	4.8	0	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	37.6	22.8	35.9	23.2	34.0	23.2	25.5	32.4	24.9	31.7	24.2	30.9	6.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.7	28.3	22.9	18.6	27.8	22.4	18.0	27.4	80.8	32.3	77.8	31.9	75.3	30.9	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.3	11.2	10.1	-16.8	39.3	2.6	1.8	-18.7	40.7	-20.2	41.7	-21.7	42.7	-23.6	44.1		
			-16.3	26.9	3.5	1.3	-18.8	27.8	-20.9	28.6	-22.9	29.3	-25.4	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	1.0	3.4	8.9	13.8	19.5	24.9	27.1	26.5	22.4	15.4	8.4	2.6
	DBStd	10.47	6.08	6.47	6.43	5.08	4.46	3.46	3.15	3.21	4.34	5.06	5.43	5.43
	HDD10.0	938	282	198	97	22	1	0	0	0	1	13	91	234
	HDD18.3	2359	537	417	296	150	42	2	0	0	13	117	299	487
	CDD10.0	2601	4	14	62	136	295	447	531	513	371	181	43	6
	CDD18.3	981	0	0	4	13	77	198	272	254	134	26	1	0
	CDH23.3	10383	0	1	34	136	668	2066	3166	2804	1278	221	8	0
	CDH26.7	4842	0	0	5	24	218	946	1631	1411	551	56	1	0
Wind	WSAvg	5.0	4.9	5.1	5.7	5.8	5.1	5.0	4.4	4.3	4.6	4.8	5.1	4.8
Precipitation	PrecAvg	917	20	36	63	83	140	151	106	105	74	73	38	29
	PrecMax	1327	67	103	141	196	294	311	238	243	265	205	135	87
	PrecMin	645	1	2	12	15	54	52	12	13	8	15	0	2
	PrecStd	189	17	25	36	44	65	68	56	58	64	46	39	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	22.5	26.7	29.3	33.8	37.8	39.7	40.1	37.3	31.0	24.2	18.9
		MCWB	11.5	12.9	16.2	18.3	21.1	23.1	22.7	22.3	21.4	19.8	14.9	12.4
	2%	DB	16.1	19.5	23.8	27.0	31.2	35.5	37.5	37.5	33.9	28.0	22.0	16.2
		MCWB	9.3	11.5	15.4	17.3	21.0	23.5	23.2	22.5	22.0	18.0	14.8	10.5
	5%	DB	13.0	16.3	21.4	24.8	29.1	33.6	35.9	35.4	32.2	25.9	19.2	13.4
		MCWB	7.3	10.1	14.3	16.3	20.6	23.4	23.6	23.1	21.6	17.5	13.0	8.7
	10%	DB	10.2	13.5	18.9	22.5	27.2	31.9	33.9	33.2	30.0	23.4	17.0	11.0
		MCWB	5.8	8.5	13.1	15.4	20.2	22.9	23.5	23.1	20.8	16.9	11.9	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	15.6	19.9	20.8	24.1	26.2	26.7	26.3	24.6	22.6	18.0	15.6
		MCDB	16.2	18.1	22.6	24.8	29.0	32.9	32.8	33.1	31.9	26.6	21.0	17.5
	2%	WB	10.4	13.1	17.9	19.5	23.0	25.2	25.5	25.3	23.4	21.2	15.8	12.2
		MCDB	14.5	16.9	21.3	23.9	28.0	32.0	32.4	32.2	30.3	24.9	19.4	14.2
	5%	WB	8.0	11.2	15.7	18.3	22.2	24.5	24.9	24.6	22.6	19.7	14.3	9.1
		MCDB	12.2	15.8	19.8	22.7	27.0	30.9	31.9	31.4	29.1	23.3	17.9	12.4
	10%	WB	6.0	8.5	13.6	16.7	21.3	23.7	24.2	23.8	21.9	18.0	12.6	7.0
		MCDB	10.1	13.6	17.9	21.4	25.8	29.8	31.1	30.5	27.9	21.8	16.4	10.4
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.4	11.9	11.7	10.9	11.0	11.2	11.4	11.5	11.7	11.3	10.2
		MCDBR	15.4	15.6	15.2	14.6	12.9	13.1	13.6	14.1	13.6	14.1	14.3	13.9
		MCWBR	9.7	9.7	8.5	8.1	5.6	4.7	3.8	4.1	4.6	6.7	8.2	9.1
	5% WB	MCDBR	14.1	13.8	12.2	12.3	11.0	11.7	11.6	11.5	11.7	10.8	11.8	12.4
		MCWBR	9.6	9.8	8.2	8.1	6.0	5.1	4.7	4.6	4.7	6.6	8.4	9.6
Clear-Sky Solar Irradiance	taub	0.287	0.298	0.333	0.373	0.408	0.413	0.421	0.425	0.380	0.337	0.298	0.280	
	taud	2.498	2.477	2.408	2.319	2.296	2.312	2.329	2.272	2.374	2.484	2.524	2.526	
	Ebn at Noon	907	939	931	906	876	868	857	846	869	881	887	893	
	Edh at Noon	79	91	108	126	131	129	126	130	111	90	76	72	
All-Sky Solar Radiation	RadAvg	2.52	3.25	4.24	5.33	5.96	6.81	6.71	6.02	5.00	3.71	2.73	2.16	
	RadStd	0.25	0.47	0.28	0.36	0.53	0.50	0.41	0.34	0.36	0.42	0.28	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page